

A Hybrid Deep Learning Approach With Explainable AI for Diabetic Retinopathy Classification

Critical Review and Technical Evaluation

Research Review Group

Technical Writing / Research Paper Review

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Publication

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Study

- Dataset: APTOS 2019
- Images: 3,662
- Task: 5-class DR classification
- CNNs + Ensemble + Hybrid
- Explainability using Grad-CAM

Main Focus: Performance + Interpretability + Computational Analysis

Diabetic Retinopathy

Diabetic Retinopathy (DR) is a diabetes-related retinal condition that can cause vision impairment and blindness, particularly when it progresses without timely detection and treatment.



- Manual retinal screening requires trained professionals.
- Large-scale screening can be time-consuming.
- Automated image classification can assist the screening process.
- The paper investigates five-class severity classification.

Research Problem and Motivation

Class Imbalance

APTOS 2019 contains substantially more No DR images than severe DR categories.

CNN Limitations

Different CNN architectures learn different visual representations of retinal features.

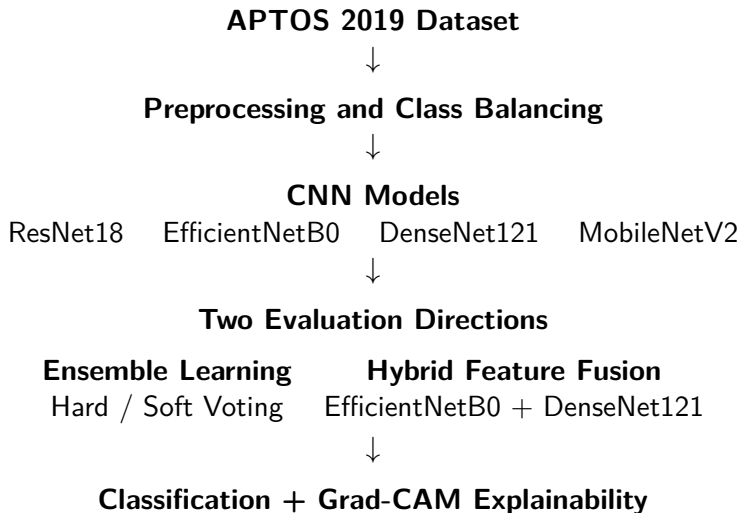
Explainability

High classification performance alone does not explain where the model is focusing.

Paper's Direction

Class Balancing + Multiple CNNs + Ensemble / Hybrid + Grad-CAM

Proposed Research Framework



Dataset and Preprocessing

APTOS 2019

- Total images: 3,662
- Five DR severity classes
- Training: 2,562
- Validation: 733
- Testing: 367

Preprocessing

- Images resized to 224×224
- ImageNet mean/std normalization
- Data augmentation
- Stratified image-level splitting

Class	Approximate Share
No DR	49%
Moderate DR	27%
Mild DR	10%
Proliferative DR	8%
Severe DR	5%

SMOTE Class Balancing: Critical Analysis

Authors' Approach

- SMOTE used to address class imbalance.
- Applied to flattened normalized pixel representations.
- Training and validation subsets were balanced.
- Test set remained untouched.

Our Critical Observation

SMOTE is traditionally designed for feature vectors.

Pixel-level interpolation in high-dimensional retinal images may not necessarily correspond to anatomically realistic retinal structures.

This motivates comparison with feature-level oversampling, class-weighted loss, or other imbalance strategies.

Key Question: Does pixel-level synthetic data preserve clinically meaningful patterns?

CNN Backbones Evaluated

Architecture	Parameters	Main Feature	Purpose
ResNet18	11.18M	Residual connections	Baseline
EfficientNetB0	4.01M	Compound scaling	Efficient representation
DenseNet121	6.96M	Dense feature reuse	Feature richness
MobileNetV2	2.23M	Inverted bottlenecks	Lightweight deployment

Reason for Comparison

The architectures represent different trade-offs between feature representation, model size, and computational complexity.

Ensemble Learning vs. Hybrid Fusion

Ensemble Learning

Hard Voting

Each CNN produces a class label. The majority class becomes the final prediction.

Soft Voting

The class probability distributions from the models are aggregated.

Hybrid Feature Fusion

EfficientNetB0:

1280-D

DenseNet121:

1024-D

Concatenation:

$$1280 + 1024 = 2304\text{-D}$$

Classifier:

$$2304 \rightarrow 512 \rightarrow 5$$

Experimental Setup

Training

- Framework: PyTorch
- Optimizer: Adam
- Learning rate: 1×10^{-4}
- Weight decay: 1×10^{-5}
- Batch size: 32
- Training epochs: 20

Evaluation Metrics

- Accuracy
- Precision
- Recall
- F1-score
- Class-wise ROC-AUC
- Confusion matrix
- Computational complexity
- Grad-CAM

Hardware

Intel i7-14700 CPU, 32 GB RAM and NVIDIA RTX 3060 GPU with 16 GB VRAM.

Experimental Performance

Model	Accuracy	F1-score
ResNet18	85.01%	84.39%
EfficientNetB0	83.65%	82.81%
DenseNet121	82.83%	82.12%
MobileNetV2	83.38%	82.95%
Ensemble	85.29%	84.58%
Hybrid	84.02%	82.91%

85.29%
Ensemble

85.01%
ResNet18

84.02%
Hybrid

Key Result: Hybrid vs. Ensemble vs. ResNet18

Reported Accuracy

Model	Accuracy
Ensemble	85.29%
ResNet18	85.01%
Hybrid	84.02%

Critical Observation

The proposed hybrid architecture does not achieve the highest overall accuracy among the evaluated approaches.

- The ensemble provides the highest reported accuracy.
- ResNet18 slightly exceeds the hybrid in overall accuracy.
- The hybrid was primarily designed to investigate complementary feature fusion.

Computational Complexity and Deployability

Model	Parameters	GFLOPs
MobileNetV2	2.23M	0.33
EfficientNetB0	4.01M	0.41
DenseNet121	6.96M	2.90
ResNet18	11.18M	1.82
Hybrid	14.45M	19.86

Observation

The hybrid combines two feature extractors and therefore has substantially greater computational complexity.

Important Limitation

The paper states that explicit inference latency for the hybrid model was not independently benchmarked.

Configuration	Feature Source	Purpose
DenseNet121	Dense features	Baseline
EfficientNetB0	Multi-scale features	Baseline
Hybrid	Both feature sets	Fusion analysis

Finding

The paper reports that the hybrid configuration improves accuracy and macro F1 compared with its individual DenseNet121 and EfficientNetB0 components.

Interpretation: Feature fusion provides complementary representations.

Explainable AI: Grad-CAM

Grad-CAM Process

- 1 Input retinal image
- 2 Model prediction
- 3 Calculate gradients
- 4 Generate activation map
- 5 Resize and overlay heatmap

Purpose

Grad-CAM is used to visualize regions of the retinal image that contribute strongly to the model's prediction.

The hybrid model is analyzed separately through its EfficientNetB0 and DenseNet121 branches.

Critical Perspective

A visually plausible heatmap demonstrates where the model is attending, but does not by itself prove that the model's reasoning is clinically correct.

Quantitative Grad-CAM Analysis

Model / Branch	Concentration Score
EfficientNetB0	0.6586
DenseNet121	0.4244

Paper's Finding

The paper reports that EfficientNetB0 demonstrates more spatially concentrated activation than DenseNet121. Concentration scores were calculated over 50 test images.

Our Critical Interpretation

Higher concentration indicates more localized attention according to the proposed metric. However, concentration alone does not establish clinical correctness of the highlighted region.

Critical Evaluation of the Paper

Strengths

- Multiple CNN baselines
- Hard and soft ensembles
- Hybrid feature fusion
- Ablation study
- Multiple evaluation metrics
- Quantitative Grad-CAM

Limitations

- Single dataset
- No external validation
- Patient identifiers unavailable
- Pixel-space SMOTE
- High hybrid complexity
- No independent hybrid latency benchmark

Questions Raised

- Are synthetic pixel samples realistic?
- Does concentration imply correctness?
- Is the computational cost justified?
- How well does it generalize?

Recommended Improvements

- 1 External validation on independent datasets.
- 2 Compare alternative imbalance-handling strategies.
- 3 Investigate pruning, quantization and knowledge distillation.
- 4 Validate XAI results with additional explainability methods and domain experts.
- 5 Explore uncertainty-aware predictions.

Our Assessment

Contribution

The study combines CNN comparison, ensemble learning, feature fusion and quantitative explainability.

Evidence

The internal evaluation is broad, but it relies on APTOS 2019.

Main Concern

Generalization and the validity of pixel-space SMOTE require further investigation.

Key Takeaways

1. Predictive Performance

The ensemble achieved the highest reported accuracy of **85.29%**.

2. Hybrid Architecture

The hybrid achieved **84.02%** while requiring **14.45M parameters** and **19.86 GFLOPs**.

3. Explainability

Quantitative Grad-CAM adds a numerical analysis of spatial attention beyond conventional visual heatmaps.

4. Generalization

External validation is still needed to determine robustness across different datasets, imaging devices and populations.

Thank You

Questions and Discussion

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